



WorldCereal

Face2Face Training on WorldCereal crop mapping system in Tunisia

May 5th –7th, 2026

In partnership with OSS



1. Purpose and scope

1.1 Background on the WorldCereal processing system

The **WorldCereal system** represents an open-source, flexible and scalable processing system allowing anyone to generate **customized crop maps** for their area, season and crop types of interest. To accomplish this, the system leverages public and/or user-private reference data, free and open high-resolution Earth Observation (EO) imagery, meteorological time series and the latest innovations in the field of deep learning and machine learning.

The processing system by default is implemented on the **Copernicus Data Space Ecosystem (CDSE)**, requiring no local installation or deployment. The different components of the system can be operated through a set of user-friendly web interfaces and Jupyter notebooks within pre-installed Python environments. Processing credits are partially provided for free through CDSE or can be purchased with or without ESA Network of Resources (NoR) sponsorship.

Through the WorldCereal system, you can:

- **Prepare and harmonize your crop reference data** to make it ready for use by our classification model training pipelines, ensuring full compatibility with our extensive public database of reference data
- **Fetch the relevant EO and ancillary data** matching your reference data and derive meaningful, AI-powered embeddings to be used for model training
- **Train your customized crop identification algorithm** based on a combination of public and private reference data, for your crop types of interest
- **Generate crop maps at any scale** in an efficient way based on your customized crop model

1.2 Description and objectives of the F2F training

This hands-on, technical F2F training aims to enable participants to integrate the cloud-based WorldCereal processing system into their day-to-day toolset for local to global EO-based crop monitoring. Together with the instructors, participants will go through all the steps involved in tailored crop mapping, using their own reference data. User-provided reference data is by default treated as private data in the WorldCereal system, yet participants can choose to share with the WorldCereal consortium. Relevant concepts will be explained where needed, but the main focus will be on working with the system in a practical way.

By the end of the training, participants will be able to:

- Use our data collection application to collect relevant and suitable in-situ reference data for EO-based crop mapping purposes
- Select, review and prepare in-situ reference data to train crop identification algorithms
- Access and explore public reference data on crop types available in the WorldCereal system
- Train custom crop type models tailored to your area and crops of interest
- Deploy and run custom crop type models anywhere, ensuring scalability and flexibility

2. Date and Location

Training dates - May 5th – May 7th 2026

Location - Tunis, Tunisia

Address - OSS premises (BP 31 Bd du Leader Yasser Arafet, Tunis 1080, Tunisia)

3. Detailed schedule

A preliminary schedule is provided in the table below. Adaptations to the schedule can be made depending on specific needs of participants.

Day	Ses sion	Time	Subject	Responsible Partner
1		8.30 - 9.00	Registration	OSS
	1.1	9.00 – 9.20	Welcome by OSS	OSS
	1.1	9.20 – 9.40	Welcome by Ministry of Agriculture	Ministry of Agriculture
	1.1	9.40 – 10.00	Introductions - Round table	All
	1.2	10.00 – 10.30	Introduction to Worldcereal for crop mapping and its applications	VITO
		10.30 – 11.00	<i>Coffee break</i>	OSS
	1.3	11.00 – 11.30	The WorldCereal Processing Hub – generating cropland masks	VITO
	1.4	11.30 – 12.30	Reference data: definition, collection and harmonization	VITO
		12.30 – 13.30	<i>Lunch break</i>	OSS
	1.5	13.30 - 14.30	Reference Data Exploration through WorldCereal Reference Data Module	VITO
	1.6	14.30 – 17.00	Field campaign showcasing data collection	OSS
2	2.1	9.00 – 10.30	Adding and preparing reference data for crop mapping (upload + extractions)	VITO
		10.30 - 11.00	<i>Coffee break</i>	OSS
	2.2	11.00 - 12.30	Generating crop type maps using the WorldCereal system	VITO
		12.30 – 13.30	<i>Lunch break</i>	OSS
	2.2	13.30 – 15.00	Generating crop type maps using the WorldCereal system	VITO
		15.00 – 15.30	<i>Coffee break</i>	OSS
	2.3	15.30 – 17.00	Preparing presentations day 3	All

3	3.1	9.00 – 10.00	Crop map validation and crop statistics	VITO/IIASA
	3.2	10.00 – 10.30	Dedicated time to work on presentations	All
		10.30 - 11.00	Coffee break	OSS
	3.3	11.00 – 12.30	Participant presentations on their work and discussion on how to integrate WorldCereal in their work (Part 1)	All
		12.30 – 13.30	Lunch break	OSS
	3.3	13.30 – 15.00	Participant presentations on their work and discussion on how to integrate WorldCereal in their work (Part 2)	All
		15.00 – 15.30	Coffee break	OSS
	3.4	15.30 – 16.30	Training feedback, final discussion and closing	All

4. More details on specific sessions

Day 1

Session 1.2: We will start with an introduction to crop mapping from space, covering open-source satellite constellations, ground truth data collection, and the key tools and algorithms used for data processing. This will be followed by an overview of the WorldCereal System, guiding participants through the motivation behind its development and the evolution of the WorldCereal program.

Session 1.3: Hands-on session to showcase how anyone can generate cropland maps using the default WorldCereal model and a convenient user interface.

Session 1.4: Presentation on definition, collection and harmonization of reference data, with specific focus on how to ensure high quality reference data.

Session 1.5: Hands-on exploration on how reference data is organized in the WorldCereal Reference data module.

Session 1.6: Excursion to nearby fields for a hands-on session on collecting in-situ data.

Day 2

Session 2.1: Adding custom data to the Reference Data Module (RDM): Hands-on sessions on the customization aspects of the system including privacy settings and on how to extract training data (EO time series) based on user-provided reference data.

Session 2.2: Hands-on session on generating crop type products using the WorldCereal system. Here we will generate a crop mask for a custom region of interest using the globally trained default crop/no-crop model and make use of publicly available reference data to train and apply a customized crop type model for a region, season and crop types of interest. This will result in a deep understanding of the different steps required to operate the WorldCereal system.

Session 2.3: Participants can integrate their crop mapping results into a short presentation showcasing how they would use the WorldCereal system in their work.

Day 3

Session 3.1: Presentation on what to do next when you have created your crop map, including important considerations on map validation and translation of maps into official crop statistics.

Session 3.3: There will be presentations by different participants which will include a. Introduction of the organization b. The use of crop type mapping in their work c. Current techniques and tools used to perform the analysis d. Analysis of the crop mapping results from day 2

Session 3.4: Discussions on the workshop, learning and closing ceremony.

5. Requirements for participation

This session will be targeted towards anyone interested in learning the required technical skills to generate crop type maps based on free and open Earth Observation data using cloud-based processing infrastructure.

We expect basic prior knowledge on the concept of Earth Observation and practical experience in handling geospatial data (raster and vector files) in GIS software (QGIS or ArcGIS). Prior experience in the use of Jupyter Notebooks may come in handy, but is not required to be able to participate in this session.

Participants are required to bring their own laptop with GIS software (QGIS or ArcGIS) installed.

To guarantee a smooth experience and to get access to the demo environment (where everything will be set-up and ready to go), participants are **required to register (for free!) on the following platforms before entering the workshop:**

- Copernicus Data Space Ecosystem (CDSE) → <https://dataspace.copernicus.eu/>
- VITO's Terrascope platform → <https://terrascope.be/en>

Participants are encouraged to use their own crop reference data during the training. Reference data should be brought in the format of Shapefiles or GeoPackage files, containing either points or polygons.

6. Practical considerations

Face2Face training is typically organized for a maximum number of 20-25 participants. The training will be facilitated by one WorldCereal instructor on site. Remote participation can be supported, yet no dedicated hands-on support will be provided for online participants.

Travel and accommodation of instructors is funded by the WorldCereal project. No additional funds are available to support travel of participants, catering services during the training or renting of a training room.

7. Training room requirements

- **Room capacity:** Comfortably host up to 30 participants.
- **Equipment:** A projector is essential for presentations and screen sharing.
- **Internet:** A stable connection for all participants.
- **Video Conferencing:** Equipment for remote participation if desired by local organizers (minimum: a decent audio system; camera is optional).
- **Technical Support:** A crew on standby to resolve any hardware issues.
- **Power:** Sufficient sockets for participants and the presenter.
- **Catering:** Smooth handling of lunch and coffee/tea breaks.